Correlation of Sunrise and Sunset Variations and Their Influence on Day Length

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An exploratory analysis of sunrise and sunset timings from sunrise-and-sunset.com, focusing on their interrelationship and the associated variations in day length.

Date: 2026-06-05

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 224 entries, 0 to 226
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        224 non-null    object
 1   Country     224 non-null    object
 2   Sunrise     224 non-null    datetime64[ns]
 3   Sunset      224 non-null    datetime64[ns]
 4   Day length  224 non-null    datetime64[ns]
dtypes: datetime64[ns](3), object(2)
memory usage: 10.5+ KB
```

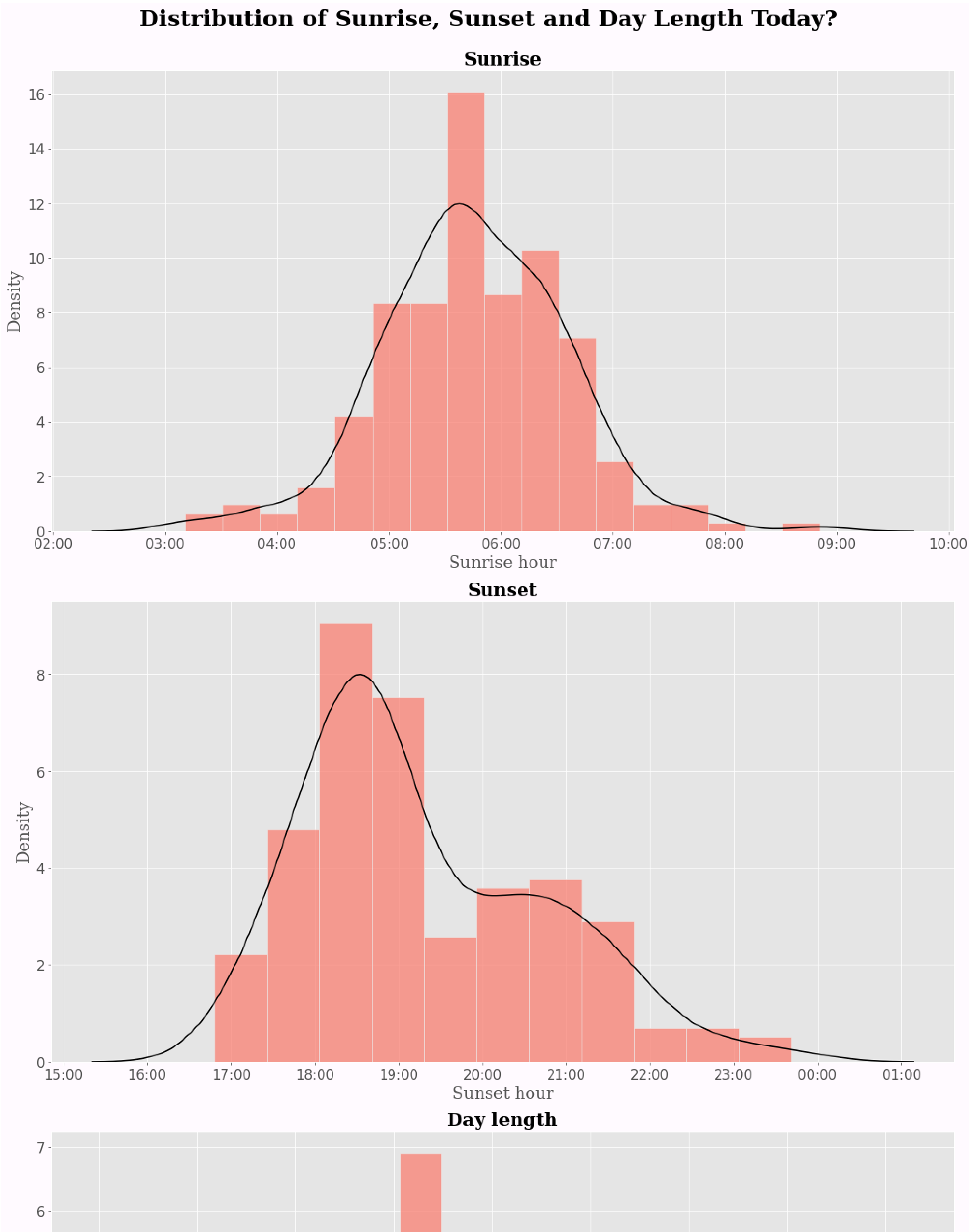
Top 5 Cities with the Shortest Days

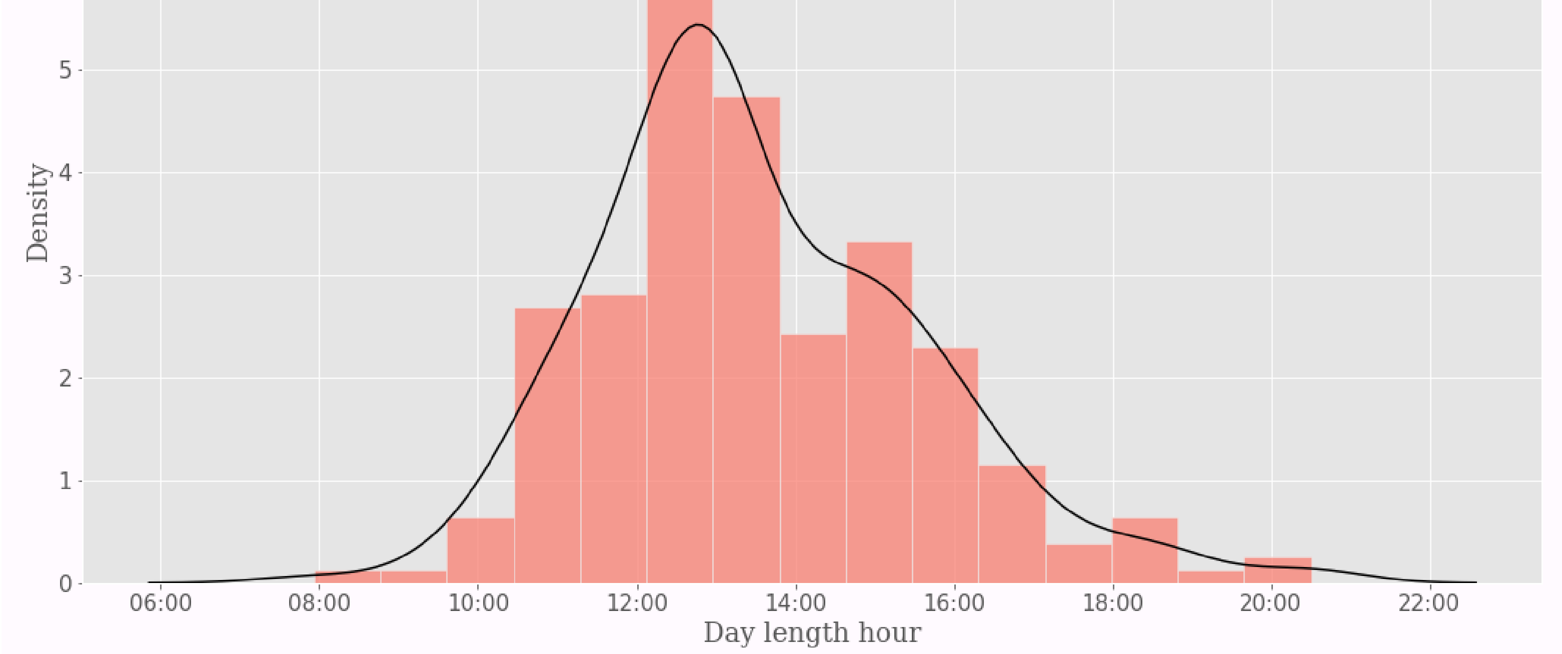
	City	Country	Sunrise	Sunset	Day length
0	Stanley	Falkland Islands	08:51:00	16:48:00	07:56:00
1	Wellington	New Zealand	07:40:00	16:57:00	09:17:00
2	Canberra	Australia	07:06:00	16:57:00	09:50:00
3	Montevideo	Uruguay	07:47:00	17:39:00	09:52:00
4	Buenos Aires	Argentina	07:55:00	17:48:00	09:53:00

Top 5 Cities with the Longest Days

	City	Country	Sunrise	Sunset	Day length
0	Mariehamn	Åland	04:23:00	22:55:00	18:31:00
1	Helsinki	Finland	04:02:00	22:35:00	18:32:00
2	Tórshavn	Faroe Islands	03:47:00	23:05:00	19:17:00
3	Reykjavík	Iceland	03:13:00	23:41:00	20:28:00
4	Nuuk	Greenland	03:11:00	23:41:00	20:30:00

Density plots (Histogram + Kernel) for Sunrise, Sunset and Day Length

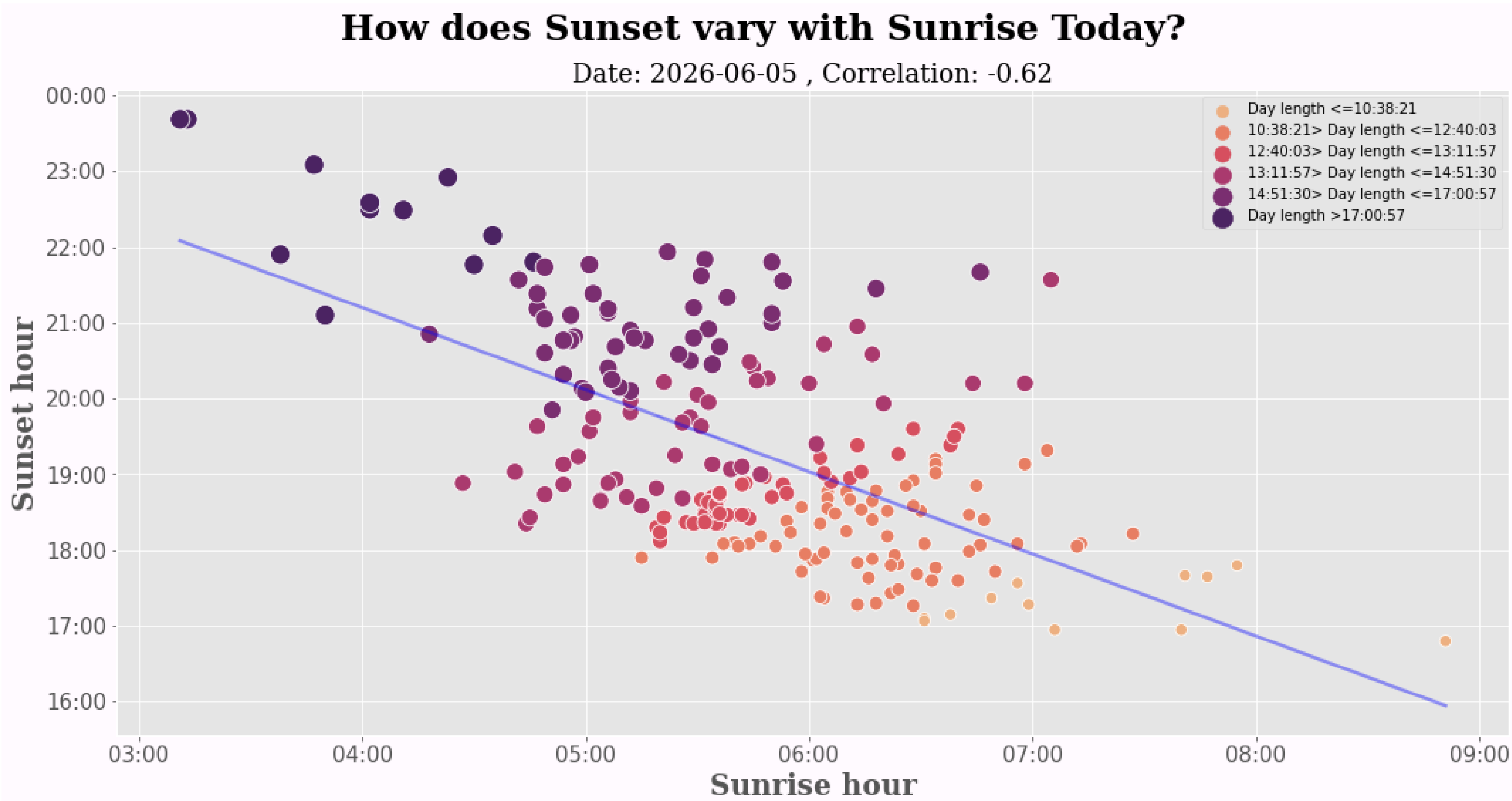




The visualizations above illustrate today's sunrise and sunset times, along with the total day length. It would be insightful to compare these results with the solar data from our respective current locations.

Scatter plot and Regression Line for Sunrise and Sunset

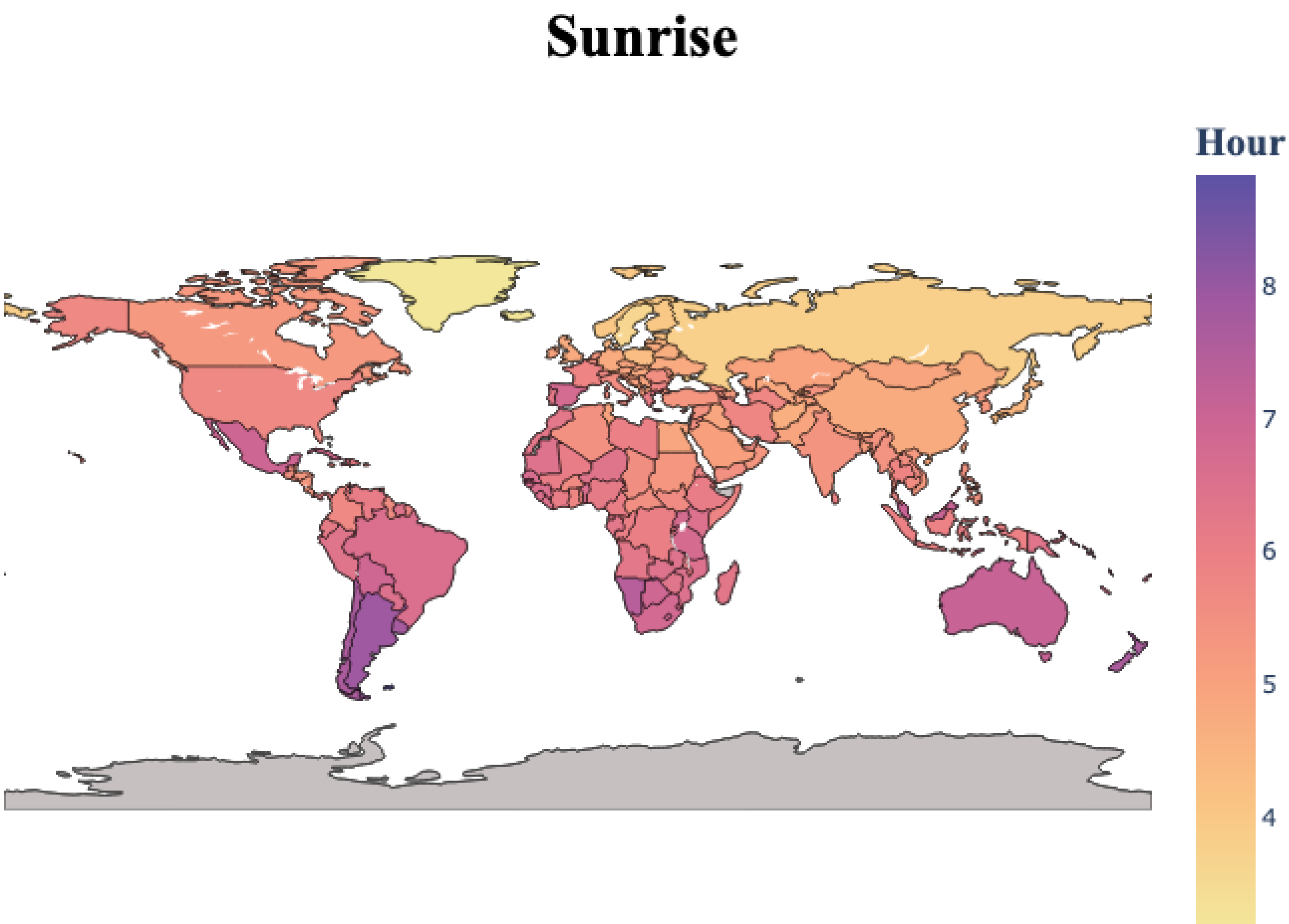
Let's explore how sunrise and sunset timings vary with each other



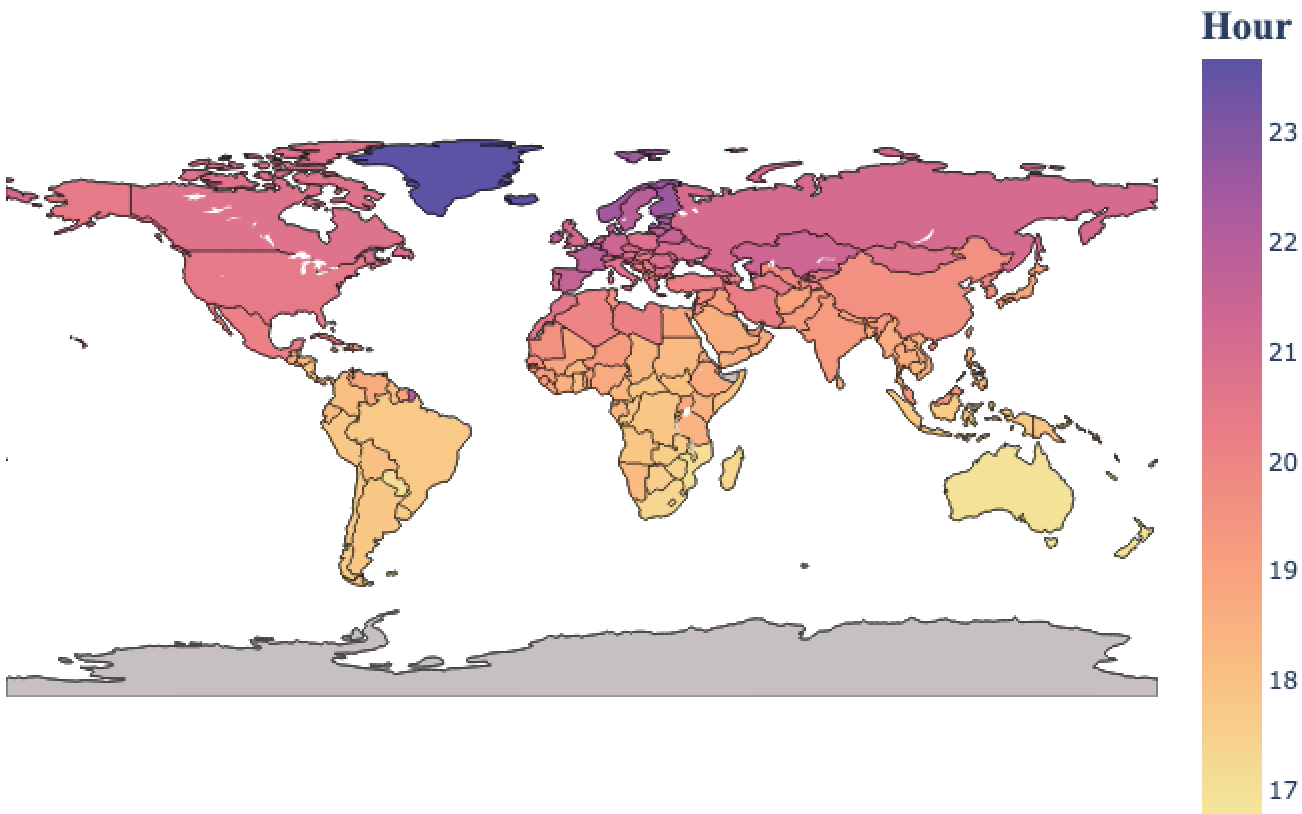
The analysis reveals a clear inverse relationship between sunrise and sunset times: later sunrises correlate with earlier sunsets and are linked with shorter day lengths.

In the visualization, cities marked in **deep purple** represent earlier sunrises and later sunsets (the longest days). Conversely, cities marked in **light orange** represent later sunrises and earlier sunsets, corresponding to the shortest day lengths in the dataset.

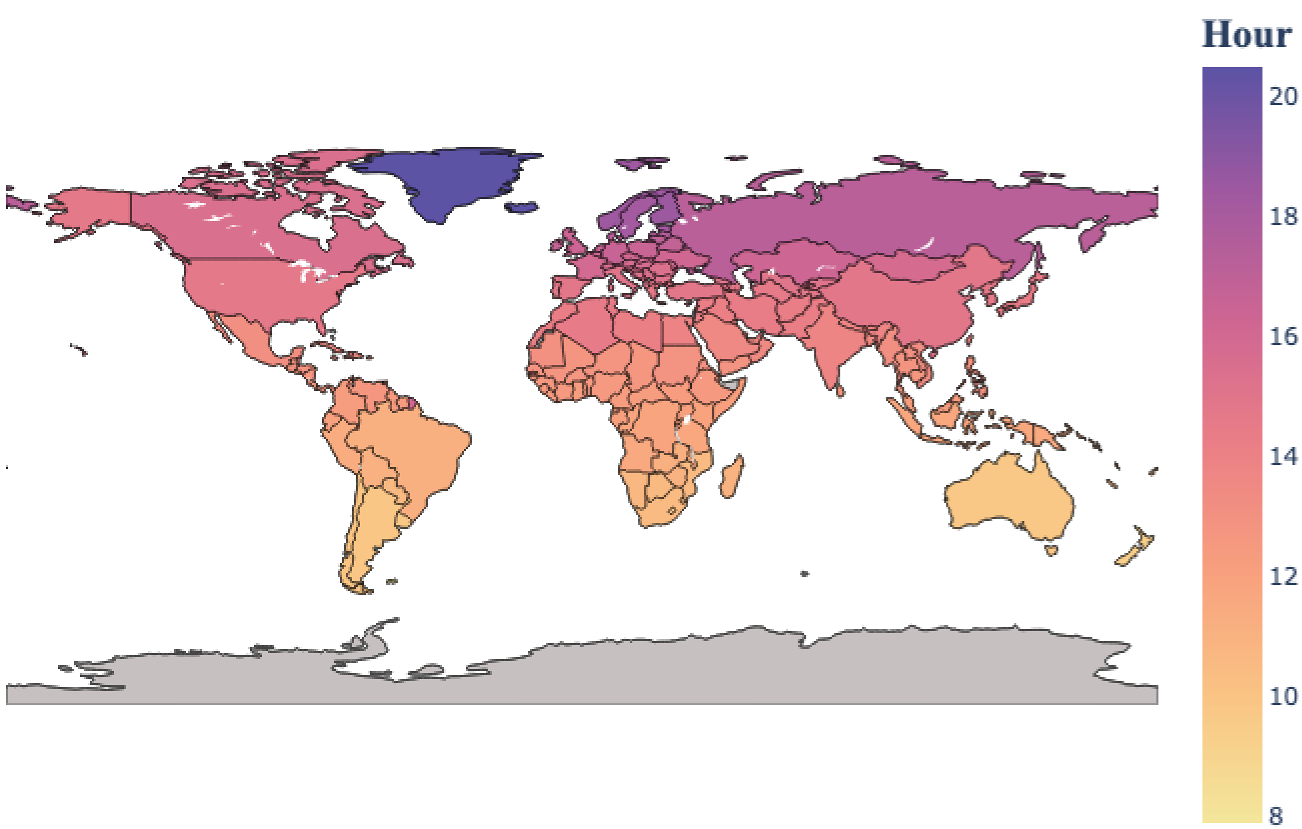
Plot choropleth world maps for Sunrise, Sunset and Daylengths



Sunset



Day length



The maps above illustrate the global distribution of sunrise, sunset, and day length. How does the data for your country compare to these global trends?